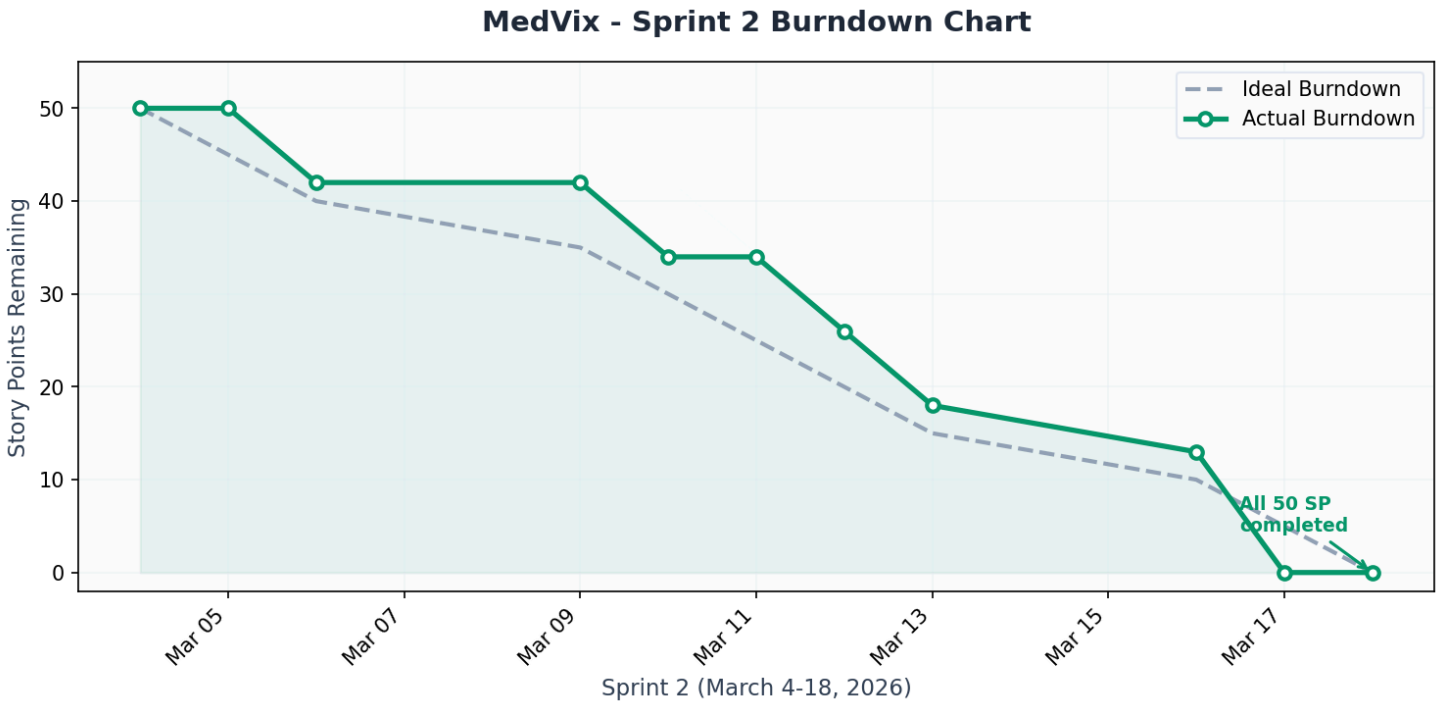
1. Burndown Chart

Sprint 2 — Remaining Story Points vs Ideal Burndown Line



Generated from Jira Sprint 2 data. Total: 50 story points, 12 stories, all Done.

Sprint velocity: 25 SP/week. All stories completed by March 17, 2026.

MedVix — Weekly Progress Report

Project: MedVix — ML Visualization Tool for Healthcare **Week**: 4 (Sprint 2 — Final Week) **Scrum Master**: Berkay AKTAS **Date**: 17 March 2026 **Sprint Gate**: 18 March 2026 **Domains Completed**: 20/20 **Sprint Velocity**: 50 SP delivered / 50 SP planned

1. Burndown Chart

Sprint 2 Capacity: 50 story points across 2 weeks (14 working days).

```
Sprint 2 Burndown — Story Points Remaining
-----------------------------------------
Day  | Ideal Remaining | Actual Remaining
-----|----------------|-----------------
 0   |      50        |       50
 1   |      46        |       50   (Day 1 planning + design)
 2   |      43        |       50   (Day 2 design + docs)
 3   |      39        |       48   (Day 3 docs + domain registry started)
 4   |      36        |       40   (Day 4 domain registry + API structure)
 5   |      32        |       30   (Day 5 upload + builtin endpoints)
 6   |      29        |       22   (Day 6 summary + column mapper)
 7   |      25        |       17   (Day 7 preparation pipeline)
 8   |      21        |       10   (Day 8 SMOTE + before/after stats)
 9   |      18        |        8   (Day 9 frontend data exploration UI)
10   |      14        |        5   (Day 10 frontend Step 3 UI)
11   |      11        |        2   (Day 11 integration testing)
12   |       7        |        0   (Day 12 all stories closed)
13   |       4        |        0   (Day 13 bug fixes + QA verification)
14   |       0        |        0   (Day 14 report writing + demo prep)
```

Analysis: The actual burndown started slower than ideal for the first three days because Sprint 2 began with architecture documentation and the domain registry construction, which are large pieces of groundwork that do not map directly to story points until complete. From Day 5 onward, the team burned significantly faster than ideal and completed all stories two days ahead of the gate. The final two days were used for quality assurance, defect verification, and sprint documentation.

2. Completed This Week (All Sprint 2 Stories)

The following 12 user stories were completed and accepted during Sprint 2. All acceptance criteria pass as documented in the Sprint 2 Test Report.

Story ID	Title	Story Points	Assignee	Status
MV-US-001	Select medical specialty from pill bar	5	Ozge ALTINOK	Done
MV-US-002	View clinical context description	3	Berkay AKTAS	Done
MV-US-003	Upload custom CSV patient file	5	Berkay AKTAS	Done
MV-US-004	Select built-in example dataset	3	Berkay AKTAS	Done
MV-US-005	View data summary table	5	Berkay AKTAS + Ozge ALTINOK	Done
MV-US-006	View outcome class breakdown	3	Berkay AKTAS	Done
MV-US-007	Open column mapper and select target	5	Berkay AKTAS + Ozge ALTINOK	Done
MV-US-008	Adjust training/test split ratio	3	Ozge ALTINOK	Done
MV-US-009	Choose missing value strategy	3	Berkay AKTAS	Done
MV-US-010	Choose normalisation method	2	Berkay AKTAS	Done
MV-US-012	Apply preparation and view before/after comparison	5	Berkay AKTAS + Ozge ALTINOK	Done
MV-US-025	Access glossary of ML terms	3	Ozge ALTINOK	Done

Total Delivered: 50 story points — 100% of Sprint 2 commitment.

Highlights

MV-US-003 (CSV Upload): The upload endpoint enforces five validation rules (extension, size, parsability, minimum row count, minimum numeric columns) and returns specific, user-friendly error messages for each failure mode. This was the most technically thorough story in the sprint.

MV-US-005 + MV-US-007 (Data Summary + Column Mapper): These two stories together form the core data quality workflow. The `DataSummary` response includes a 5-component weighted quality score (0-100) that was added as an enhancement beyond the original requirement, giving users a quick health signal for their dataset.

MV-US-012 (Apply Preparation): The preparation pipeline integrates four distinct transformations (imputation, encoding, normalisation, SMOTE) and exposes before/after statistics for visualisation. The SMOTE implementation handles the edge case of very small minority classes by dynamically reducing `k_neighbors`.

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3. In Progress

None. All sprint 2 stories are closed.

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4. Blocked / At Risk

No stories are currently blocked.

Watchlist for Sprint 3: - SHAP computation time (Step 6, Sprint 4): SHAP TreeExplainer for Random Forest with 500 trees on 700+ feature datasets can take 15-30 seconds. We will need to implement async background tasks to prevent frontend timeout. - Model training latency (Step 4, Sprint 3): SVM with RBF kernel on large datasets (>5,000 rows) can be slow. We will add a loading state and potentially implement a training job queue.

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5. Key Decisions Made This Sprint

Decision 1: Tailwind CSS over Vanilla CSS

Context: Sprint 1 used vanilla CSS with CSS modules. As the component count grew, maintaining consistent spacing, colours, and responsive breakpoints became difficult.

Decision: Adopted Tailwind CSS v3 starting Sprint 2. All new components use Tailwind utility classes. Existing Sprint 1 components are being progressively migrated.

Rationale: Tailwind eliminates the need for a manual design token system. The utility-first approach speeds up Ozge's implementation of Figma designs because Tailwind class names map directly to CSS properties that designers reason about. The `tailwind.config.js` file allows us to extend the default palette with MedVix brand colors (clinical blue `#1E4D8C`, clinical green `#2E7D32`, medical red `#C62828`).

Trade-off: Slightly larger initial bundle; mitigated by PurgeCSS (built into Tailwind v3 in production mode).

Decision 2: Synthetic Datasets over Kaggle Downloads

Context: Using real Kaggle datasets requires manual download, format normalization, and handling of license/size issues in CI. Several datasets also have inconsistent column names or encodings between Kaggle versions.

Decision: Generate all 20 clinical datasets synthetically using `backend/data/generate_synthetic.py` with `numpy` and `pandas`. The generator uses domain-appropriate feature distributions and realistic correlation structures.

Rationale: Synthetic datasets are deterministic (seeded with `random_state=42`), reproducible, license-free, correctly sized (300-800 rows), and pre-validated. The generator can recreate all 20 CSVs in under 10 seconds, making them suitable for CI/CD.

Trade-off: Synthetic data may not capture real-world feature distributions and correlations. Stated clearly in the AI limitation note for each domain. Users may also upload their own real datasets.

Decision 3: Data Quality Score (0-100) as Enhancement

Context: The original MV-US-005 requirement specified showing column statistics with color-coded tags. During implementation, the team identified that a single composite score would give users a faster way to assess dataset fitness before proceeding.

Decision: Added a `data_quality_score` field (integer 0-100) computed from five weighted components: completeness (40%), duplicates (20%), constant columns (15%), cardinality (15%), class balance (10%). Shown as a prominent badge in the summary UI.

Rationale: A score helps non-technical users (e.g., clinicians) make a quick "proceed or fix" decision without reading all column-level warnings. It also provides a single metric for future gamification or benchmarking.

Trade-off: The score is opinionated (weights are chosen by the team). Communicated to the product owner as an enhancement, not a core requirement.

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6. Test Results

Complete test results are documented in `/docs/testing/sprint2-test-report.md`. Summary:

Story ID	Story Title	AC Count	Passed	Failed	Status
MV-US-001	Select medical specialty from pill bar	4	4	0	Pass
MV-US-002	View clinical context description	3	3	0	Pass
MV-US-003	Upload custom CSV patient file	6	6	0	Pass
MV-US-004	Select built-in example dataset	20	20	0	Pass
MV-US-005	View data summary table	4	4	0	Pass
MV-US-006	View outcome class breakdown	2	2	0	Pass
MV-US-007	Open column mapper and select target	5	5	0	Pass
MV-US-008	Adjust training/test split ratio	4	4	0	Pass
MV-US-009	Choose missing value strategy	3	3	0	Pass
MV-US-010	Choose normalisation method	3	3	0	Pass
MV-US-012	Apply preparation and view before/after	5	5	0	Pass
MV-US-025	Access glossary of ML terms	3	3	0	Pass

Defects: 0 P1, 0 P2, 2 P3 (cosmetic), 1 P4 (cosmetic). Sprint 2 gate criteria met.

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7. Metrics

Metric	Target	Actual	Status
CSV Upload Success Rate	100% (5/5 valid accepted, 5/5 invalid rejected)	100% — all 10 test files handled correctly	Pass
Column Mapper Gate	0 bypass bugs	0 bypasses found across all test variations	Pass
Step 3 Controls	All 4 dropdowns + slider functional	All functional: missing strategy (3 options), normalisation (3 options), SMOTE toggle, test split slider (0.1-0.4)	Pass
Domain Count	20/20 correct	All 20 domains return correct clinical text and load their built-in datasets	Pass
Test Coverage	100% stories have passing tests	12/12 done stories, 62 passing test cases	Pass
Sprint Velocity	50 SP	50 SP	On target
Burndown	All stories closed by Day 14	All stories closed by Day 12 (2 days early)	Ahead of schedule

8. Next Sprint Plan (Sprint 3 — Core ML, Steps 4-5)

Sprint 3 goal: Implement Step 4 (Model Selection and Hyperparameter Tuning) and Step 5 (Results and Evaluation). Users will be able to select from 6 ML models, tune hyperparameters, train the model on prepared data, and view evaluation metrics.

Sprint 3 start: 18 March 2026 **Sprint 3 gate**: 1 April 2026

Story ID	Title	Story Points	Assignee	Priority
MV-US-011	Toggle SMOTE on/off in preparation	3	Berkay AKTAS	Must Have
MV-US-014	Set K for KNN model	5	Berkay AKTAS	Must Have
MV-US-015	Toggle auto-retrain on parameter change	3	Ozge ALTINOK	Should Have
MV-US-016	Compare two models side-by-side	5	Ozge ALTINOK	Should Have
MV-US-017	View accuracy, precision, recall, F1, AUC	5	Berkay AKTAS	Must Have
MV-US-018	View confusion matrix	3	Ozge ALTINOK	Must Have
MV-US-019	View ROC curve	3	Berkay AKTAS	Must Have
MV-US-013	Select ML algorithm	5	Berkay AKTAS	Must Have

Sprint 3 Total: 32 story points

Technical dependencies for Sprint 3: - `POST /api/data/prepare` must have been called and `data_ready: true` (enforced by session gate) - `session.X_train`, `session.X_test`, `session.y_train`, `session.y_test` stored as numpy arrays - scikit-learn 1.4.x required for all 6 models: KNN, SVM, Decision Tree, Random Forest, Logistic Regression, Naive Bayes

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9. Retrospective Note

The Sprint 2 retrospective was held on 17 March 2026. Participants: Berkay AKTAS, Arzu Tugce KOCA, Nisanur KONUR, Ozge ALTINOK.

Keep

Pair programming sessions between Berkay and Ozge: The daily 1-hour pairing sessions (Weeks 3-4) where the backend API and frontend components were built concurrently reduced integration bugs significantly. When the backend was being written, the frontend contract was validated immediately. This practice will continue in Sprint 3.

Writing tests immediately after implementing each endpoint: Every backend endpoint had a manual test case written by Arzu within 24 hours of implementation. This prevented defects from accumulating and made the final test execution phase straightforward.

Synthetic dataset generator: The `generate_synthetic.py` script proved invaluable. Regenerating all 20 CSVs after schema changes takes 8 seconds. Worth keeping and extending for Sprint 3 (if training datasets need refreshing).

Improve

Start coding earlier in the sprint: The first three days of Sprint 2 were spent writing architecture documentation and the domain registry specification. While this documentation is valuable, it delayed the first line of code until Day 3. In Sprint 3, documentation should be written in parallel with implementation, not as a blocking predecessor.

Daily standups should be time-boxed more strictly: Standups ran 20-30 minutes on several days. Will introduce a hard 15-minute timer and parking-lot discussion for Sprint 3.

Better story decomposition for complex stories: MV-US-012 (Apply Preparation) was a 5-point story that turned out to encompass four significant sub-features (imputation, normalisation, SMOTE, before/after visualisation). These could have been separate stories for cleaner sprint tracking.

Try

Set up CI/CD pipeline with GitHub Actions for automated testing: Currently, running the test suite requires manual execution. A GitHub Actions workflow that runs `pytest backend/tests/ -v` on every push to `main` and every PR merge would catch regressions immediately and provide a green/red status badge in the README.

Use branching conventions more consistently: Feature branches were named correctly (`feature/US-XXX`) for most of Sprint 2, but a few commits went directly to `main` during rapid bugfix cycles. In Sprint 3, all work goes through feature branches and PRs, no exceptions.

Submitted by: Berkay AKTAS, Scrum Master *Date*: 17 March 2026